

GHz and are more power efficient. Featuring the latest Adreno 540 GPU, the SoC is capable of rendering 4K UHD videos and is even compatible with Google Daydream.

The chip also comes with inbuilt support for a number of other features including eye based scanning systems (iris or retina) and better security settings.

Primary CPU	Kryo 208 octa core 4x2.45 GHz + 4x1.8 GHz
CPU architecture	ARM big.LITTLE
CPU structure	RISC
DSP type	Hexagon 682 DSP
GPU	Qualcomm Adreno 540 GPU
Year Released	2016
Devices	Samsung Galaxy S8

Snapdragon 820 and 821

The current generation of SoCs by Qualcomm is the Snapdragon 820. This chip is the one that powers a number of flagships including HTC 10, Samsung Galaxy S7 and Note 7, Xiaomi Mi 5 and the LG G5.

It has 2 cores clocked at 2.2 GHz and the remaining 2 clocked at 1.6 GHz. The design of the chip is based on the Harvard architecture because of which the CPU has a separate L1 cache (instruction and data both) which results in slightly better performance.

They both pack quad-core processors but this does not necessarily mean that these devices are slower. The number of cores don't matter that much in smartphones since most of the applications aren't designed to optimally use all the cores of the CPU. Also there exist very few resource intensive and demanding applications that will benefit from additional cores.

In fact, the high powered quad core processor, couples with the Adreno 530 GPU makes for a formidable duo, leading to the Qualcomm 820 getting the highest Antutu benchmark score of 136,383.



The Galaxy S8 is powered by the Snapdragon 835.